

WhatsApp Group Analyzer

Transform WhatsApp chat exports into actionable community intelligence — 100% free and open-source.

Python 3.9+

Streamlit 1.30+

License MIT

Cost \$0

Quick Start

```
# 1. Clone / download the project
cd wa-analyzer

# 2. Install dependencies (all free)
pip install -r requirements.txt

# 3. Run the app
streamlit run app.py
```

Then upload your WhatsApp chat export (.txt or .zip) and explore!

How to Export WhatsApp Chat

1. Open your WhatsApp group
2. Tap : **More** → **Export Chat** → **Without Media**
3. Save the .txt or .zip file
4. Upload it to the app

Features (10 Pages)

Page	Feature	Description
	Overview Dashboard	KPIs, trends, message volume, media breakdown
	User Leaderboard	Top N selector, engagement tiers, activity heatmap
	Member Directory	Auto-extracted names, locations, LinkedIn profiles
	Temporal Analytics	Hourly/daily/weekly patterns, calendar heatmap
	Topics & Content	Word cloud, topic tracking, shared resources

Page	Feature	Description
😊	Emoji & Sentiment	Emoji leaderboard, mood indicators, humor index
🌐	Social Network	Interactive interaction graph, influencer metrics
📄	AI Chat Summary	LLM-powered summarization (plug any free API)
🔍	Search Messages	Full-text search with filters
📁	Export Center	CSV, Excel, JSON export for all data

💰 Cost: \$0

Every single component is free:

Component	Library	License	Cost
Web Framework	Streamlit	Apache 2.0	Free
Charts	Plotly	MIT	Free
Data Processing	Pandas, NumPy	BSD	Free
Word Clouds	wordcloud	MIT	Free
Network Analysis	NetworkX	BSD	Free
Emoji Processing	emoji	BSD	Free
Excel Export	openpyxl, xlswriter	MIT	Free
Hosting	Streamlit Cloud	—	Free tier

AI Summary (optional, all have free tiers):

Provider	Free Tier	How to Get
Google Gemini	15 req/min	aistudio.google.com
Groq	Generous limits	console.groq.com
Ollama (local)	Unlimited	ollama.ai
OpenAI	\$5 trial	platform.openai.com

🏗️ Architecture

```

wa-analyzer/
├── app.py                # Main entry – upload + routing
├── .streamlit/config.toml # Theme configuration
├── pages/
│   └── 1_📊_Overview.py

```

```

├── 2_🏆_Leaderboard.py
├── 3_📁_Directory.py
├── 4_🕒_Temporal.py
├── 5_💬_Topics.py
├── 6_😊_Sentiment.py
├── 7_🕸_Network.py
├── 8_📄_AI_Summary.py
├── 9_🔍_Search.py
├── 10_📤_Export.py
├── utils/
│   ├── __init__.py
│   ├── parser.py           # WhatsApp format parser
│   └── helpers.py          # Shared utilities
├── assets/
│   └── style.css           # Custom theme
├── requirements.txt
└── README.md

```

🔧 Scaling Up — Future Add-ons

The modular architecture makes it easy to add:

Feature	How	Effort
AI Summaries	Uncomment AI provider in requirements.txt, add API key	5 min
Multi-group comparison	Upload multiple files, store in DuckDB	Medium
Authentication	streamlit-authenticator package	Easy
Scheduled reports	Cron job + email via smtplib	Medium
Database backend	Replace DataFrames with DuckDB/SQLite	Medium
Docker deployment	Add Dockerfile (template below)	Easy
Webhook alerts	Add Telegram/Discord bot integration	Medium

Deploy to Streamlit Cloud (Free)

1. Push code to GitHub
2. Go to share.streamlit.io
3. Connect your repo → Deploy → Done!

📄 License

MIT — Free for personal and commercial use.